SUR3520 Measurement Science/SUR5525 Least Squares Adjustment

1. OVERVIEW

Measurement science involves methodologies for analyzing and handling errors with a focus on least squares adjustments. This course will provide an introduction to common sources and types of errors that occur in geodetic measurements, as well as proper analysis procedures to handle these errors and assess their effect on final computation results, such as point coordinates or areas. Topics include error theory, statistical distributions, hypothesis testing, law of error propagation, principle of least squares, adjustment procedures (e.g., for horizontal surveys, level networks, GPS baselines), and error ellipses.

- Spring semester, 3 credits
- 100% online
- http://elearning.ufl.edu/

Course Prerequisites:
MAC 2233, STA 2023, and SUR 3641 Survey Computations, or instructor consent

Instructor:
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Communication:
- Office hours: Mondays 5-7 p in Zoom
- Use the Canvas message/Inbox feature for fastest response

Lectures:
Tuesdays: 10:40a-11:30a (period 4), Thursdays: 10:40a-12:35p (period 4+5)
Note: Lectures are prerecorded. Links to lecture recordings will be posted on Canvas in weekly modules.

Recommended textbook:
- Ghilani, C. D. (2018). Adjustment Computations - Spatial Data Analysis (6th Ed.). New York, NY: John Wiley & Sons

Required teaching materials:
- All necessary information to succeed in the course can be found in online materials (such as lecture videos, lecture notes, Excel sheets for in-class exercises) posted on the Canvas Website. Optional reading materials are marked as such whereas all other materials are required.

Software requirements:
- Microsoft Excel is the primary software used for most topics in this course. Basic skills in Microsoft Excel are essential. Students who do not have access to Microsoft Excel are encouraged to use Microsoft Excel in UF Apps (https://info.apps.ufl.edu/).
- Geodetic software (ADJUST and StarNet) is used for selected adjustment topics. Software download instructions for this software will be provided on the course website under the Week 7 and Week 11 module, respectively. StarNet and ADJUST software run on Windows and can be installed locally. Students who do not have access to a Windows computer can contact the instructor for remote access to a server which runs StarNet and ADJUST.

2. LEARNING OUTCOMES

The course objective is to provide students with the following competencies at the completion of the course:

1. Apply statistical methods to describe the distribution and quality of geodetic measurements
2. Use error propagation to model the error of indirect measurements
3. Apply the least-squares method in geodetic software and spread sheet functions to adjust various types of measurements
4. Use error ellipses to describe geodetic network quality
5. Provide a critical interpretation of computational results and concepts

The course Website (see under Modules/Course Overview) contains an alignment table which illustrates how course activities (e.g. assignments, discussion posts, quizzes) are linked to course competencies.

3. COURSE LOGISTICS

- Throughout the semester, students will be given 10 homework assignments (graduate students: 11), 3 quizzes, 4 discussion tasks, and one field lab assignment.
- Attendance points will be given for one Q&A session.
- For each type of assessment (home assignment, quiz, discussion), a due date and time is given which is usually midnight one week after the handout
- Assignments and field labs are graded based on timeliness, correctness of computations and interpretation of numerical results, with written feedback provided by the instructor; quizzes are auto-graded based on correctness of multiple choice questions with correct answers shown after completion, and discussion items are graded based on creativity, completeness, technical correctness and the number of comments provided to peers.
- There is a 1-week turnaround for assignment, field lab, and discussion grading. Quizzes are auto-graded instantaneously in Canvas.
- This course does not offer extra credit tasks
- This course is a distance education course taught partly as prerecorded lectures and partly as live lectures using the virtual classroom software Zoom. Lecture materials can be downloaded from weekly modules on the Canvas website.

Technology Requirements:

- A computer or mobile device with high-speed internet connection
- A headset and/or microphone and speakers to participate in live sessions
- For Zoom: A supported web browser on a supported operating system (Windows, Mac OS, Linux); and minimum bandwidth. More details can be found here.

Using Zoom:

Q&A sessions and office hour meetings (per individual student requests) will be conducted with the Zoom web conferencing software. Sessions can be joined by clicking a link posted on Canvas.

Grades:

Item	*Percentage*
Homework assignments (UG: 10; G: 11)	24-30%
Quizzes (4)	6-10%
Topical discussions (2)	6-8%
Home assignment discussions (4)*	3-5%
Exams (2)	46-50%
Field Lab (1)	4-8%
Total (mid-points)	**100%**

To enhance student involvement in their educational process, the class is invited to select the weight of each assessment category from the ranges provided in the table above using Canvas. The average percentage value will then be computed for each category, normalized to sum up to 100% across all items, and used for grading.

* To earn full points for this item, share one post and one response for at least four different home assignment discussions.

Grading scale:

Grade	*Percentage*	*Grade*	*Percentage*
A	90.0-100.0	C+	73.0-74.9
A-	87.0-89.9	C	67.0-72.9
B+	85.0-86.9	C-	65.0-66.9
B	77.0-84.9	D	50.0-64.9
B-	75.0-76.9	E	0-49.9

Graduate students are required to complete an additional assignment that consists of advanced tasks relating to topics taught throughout the semester. Completing the tasks requires adjustment computations that are not part of other assignments. The additional assignment counts towards the homework assignment grade. A minimum point score is not required on this additional assignment to receive a final course grade.

For information on current UF policies for assigning grade points, see https://catalog.ufl.edu/ugrad/current/regulations/info/grades.aspx

4. COURSE CONTENT

Week	Topic	Assignment (due)	Book chapter
Week 1, Jan 14	Course introduction Types of measurements and errors	Q1 (Jan 21)	ch. 1
Week 2, Jan 21	Analysis of observations Random errors	H1 (Jan 28)	ch. 2 ch. 3
Week 3, Jan 28	Distributions used in sampling theory Confidence intervals	H2 (Feb 4)	ch. 4
Week 4, Feb 4	Hypothesis testing	H3, Q2 (Feb 11)	ch. 5
Week 5, Feb 11	Law of error propagation	Q3 (Feb 18)	ch. 6
Week 6, Feb 18	Error Propagation in distance, angle, and elevation measurements	H4, D1 (Feb 25)	ch. 7-9
Week 7, Feb 25	Principles of Least Squares Adjustment	H5, D2 (Mar 11)	ch. 11

Week 8, Mar 4	**Mar 4: Midterm exam** between 5-9 pm (2 hours)		
Week 9, Mar 11	Solving nonlinear equation systems	H6 (Mar 25)	ch. 11
Mar 17 – Mar 21	*Spring break (no classes)*		
Week 10, Mar 25	Adjustment of level networks	H7 (Apr 1), F (Apr 23)	ch. 12
Week 11, Apr 1	Adjustment of trilateration networks	H8 (Apr 8),	ch. 14
Week 12, Apr 8	Adjustment of triangulation networks Error ellipse	H9 (Apr 15)	ch. 15 ch. 19
Week 13, Apr 15	Adjustment of GNSS baselines	H10, Q4 (Apr 22) HG (Apr 25)	ch. 17
Week 14, Apr 22	No lecture. Finish field lab		
Week 15	**Apr 29: Final exam** between 5-9 pm (2 hours)		

D: Discussion, Q: Quiz, H: Home assignment (HG = Graduate), F: Field lab

Exam dates:

- Midterm exam: Tue Mar 4; exam time window: 5-9 pm; exam duration: 2 hours
- Final exam: Tue Apr 29; exam time window: 5-9 pm; exam duration: 2 hours

5. POLICIES AND REQUIREMENTS

This syllabus represents current plans and objectives for this course. As the semester progresses, changes may need to be made to accommodate timing, logistics, or to enhance learning. Such changes, communicated clearly, are not unusual and should be expected.

Attendance, late submissions and make-up requests:

- Attendance is only required for the final review Q&A session listed above.
- A 10% penalty per day will be applied to late assignments. A late submission on the due date also results in a 10% deduction. Students need to submit a late assignment as an attachment in a Canvas e-mail to the instructor. Do not attach it as a note to the submission platform.
- Assignments will not be accepted if handed in more than a week after the due date (unless specified differently).
- Students may request an extension to submitting a home assignment for a justified reason but need to do so before the submission deadline.
- Quizzes cannot be taken past the deadline
- Online discussions cannot be completed past the deadline.
- Exceptions to the late policy are only allowed per university policy.

Requirements for class attendance and make-up exams, assignments and other work are consistent with university policies that can be found at:
https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/

Computer or other hardware failures, except failure of the UF e-Learning system, will not excuse students for missing assignments. Any late submissions due to technical issues MUST be accompanied by the ticket number received from the Helpdesk when the problem was reported to them. The ticket number will document the time and date of the problem. You MUST e-mail your instructor within 24 hours of the technical difficulty if you wish to request consideration. For computer, software compatibility, or access problems call the HELP DESK phone number—352-392-HELP = 352- 392-4357.

Semester Evaluation Process:

Student assessment of instruction is an important part of efforts to improve teaching and learning. At the end of the

semester, students are expected to provide feedback on the quality of instruction in this course using a standard set of university and college criteria. Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at https://gatorevals.aa.ufl.edu/students/

Students will be notified when the evaluation period opens and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via https://ufl.bluera.com/ufl/. Summaries of course evaluation results are available to students at: https://gatorevals.aa.ufl.edu/public-results/

Communication Courtesy and Professionalism:
Just as in any professional environment, meaningful and constructive dialogue is expected in this class and requires a degree of mutual respect, willingness to listen, and tolerance of opposing points of view. Respect for individual differences and alternative viewpoints will be maintained in this class at all times. All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats, as laid out in the UF Netiquette Guide for Online Courses. Failure to do so may result in loss of participation points and/or referral to the Dean of Students' Office.

Inclusive Learning Environment:
This course embraces the University of Florida's Non-Discrimination Policy, which reads, The University shall actively promote equal opportunity policies and practices conforming to laws against discrimination. The University is committed to non-discrimination with respect to race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information and veteran status as protected under the Vietnam Era Veterans' Readjustment Assistance Act.

Academic Honesty Policy:
UF students are bound by The Honor Pledge which states "We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: "On my honor, I have neither given nor received unauthorized aid in doing this assignment." The Conduct Code specifies a number of behaviors that are in violation of this code and the possible sanctions. See the UF Conduct Code website for more information. If you have any questions or concerns, please consult with the instructor or TAs in this class.

It is assumed that you will complete all work independently in each course unless the instructor provides explicit permission for you to collaborate on course tasks (e.g. assignments, papers, quizzes, exams). It is your individual responsibility to know and comply with all university policies and procedures regarding academic integrity and the Student Honor Code. Violations of the Honor Code at the University of Florida will not be tolerated. Violations will be reported to the Dean of Students Office for consideration
of disciplinary action. For more information regarding the Student Honor Code, please see:
https://policy.ufl.edu/regulation/4-040/

University Policy on Accommodating Students with Disabilities:
Students requesting accommodation for disabilities must first register with the Dean of Students Office ((https://disability.ufl.edu/). The Dean of Students Office will provide documentation to the student who must then provide this documentation to the instructor when requesting accommodation. You must submit this documentation prior to submitting assignments or taking the quizzes or exams. Accommodations are not retroactive, therefore, students should contact the office as soon as possible in the term for which they are seeking accommodations.

Software use:
All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Privacy Statement:
Our class sessions may be audio visually recorded for students in the class to refer back and for enrolled students who are unable to attend live. Students who participate with their camera engaged or utilize a profile image are agreeing to have their video or image recorded. If you are unwilling to consent to have your profile or video image recorded, be sure to keep your camera off and do not use a profile image. Likewise, students who un-mute during class and participate orally are agreeing to have their voices recorded. If you are not willing to consent to have your voice recorded during class, you will need to keep your mute button activated and communicate exclusively using the "chat" feature, which allows students to type questions and comments live. The chat will not be recorded or shared. As in all courses, unauthorized recording and unauthorized sharing of recorded materials is prohibited.

In-class recordings:
Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal education use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor.

A “class lecture” is an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and deliver by an instructor hired or appointed by the University, or by a guest instructor, as part of a University of Florida course. A class lecture does not include lab sessions, student presentations, clinical presentation such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or guest lecturer during a class session.

6. CAMPUS HELPING RESOURCES

Whole Gator App:
The Whole Gator and website and app connects UF students with resources dedicated to supporting overall health and well-being. In addition to many of the resources below it also has strategies to practice self-care.
https://one.uf.edu/whole-gator/topics

Health and Wellness:

- U Matter, We Care: If you or someone you know is in distress, please contact umatter@ufl.edu, 352-392-1575, or visit the U Matter, We Care website to refer or report a concern and a team member will reach out to the student in distress.
- Counseling and Wellness Center: Visit the Counseling and Wellness Center website or call 352-392-1575 for information on crisis services as well as non-crisis services.
- Student Health Care Center: Call 352-392-1161 for 24/7 information to help you find the care you need, or visit the Student Health Care Center website.
- University Police Department: Visit the UF Police Department website or call 352-392-1111 (or 9-1-1 for emergencies).
- UF Health Shands Emergency Room / Trauma Center: For immediate medical care call 352-733-0111 or go to the emergency room at 1515 SW Archer Road, Gainesville, FL 32608; Visit the UF Health Emergency Room and Trauma Center website.
- GatorWell Health Promotion Services: For prevention services focused on optimal wellbeing, including Wellness Coaching for Academic Success, visit the GatorWell website or call 352-273- 4450.
- Student Success Initiative, http://studentsuccess.ufl.edu

Academic Resources:

- E-learning technical support: Contact the UF Computing Help Desk at 352-392-4357 or via e-mail at helpdesk@ufl.edu
- Career Connections Center: Reitz Union Suite 1300, 352-392-1601. Career assistance and counseling services.
- Library Support: Various ways to receive assistance with respect to using the libraries or finding resources. Call 866-281-6309 or email ask@ufl.libanswers.com for more information

- Teaching Center: 1317 Turlington Hall, 352-392-2010 or to make an appointment 352- 392-6420. General study skills and tutoring.
- Writing Studio: Daytime (9:30am-3:30pm): 2215 Turlington Hall, 352-846-1138 | Evening (5:00pm-7:00pm): 1545 W University Avenue (Library West, Rm. 339). Help brainstorming, formatting, and writing papers.

Student Complaints:

- Academic Complaints: Office of the Ombuds; Visit the Complaint Portal webpage for more information.
- Enrollment Management Complaints (Registrar, Financial Aid, Admissions): View the Student Complaint Procedure webpage for more information.
- Residential Course: https://www.ombuds.ufl.edu/complaint-portal/
- Online Course: https://pfs.tnt.aa.ufl.edu/state-authorization-status/#student-complaint